Principles of Statistics

Part II

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Principles of Statistics

0. INTRODUCTION

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and $X : \Omega \to \mathbb{R}$ a random variable with distribution function

$$F(t) = \mathbb{P}\big(\{\omega : X(\omega) \leq t\}\big) = \begin{cases} \displaystyle\int_{-\infty}^{t} f(x)\,dx, & \text{if } X \text{ is continuous with pdf } f, \\ \displaystyle\sum_{x \leq t} f(x), & \text{if } X \text{ is discrete with pmf } f. \end{cases}$$

We denote $P = \mathbb{P} \circ X^{-1}$, the law or "distribution" of X.
We observe $X_1, \ldots, X_n$, which are iid copies of X, where n is the sample size.

Def 1. A statistical model is a collection $\{f_\theta : \theta \in \Theta\}$ of pdfs or pmfs, or $\{P_\theta : \theta \in \Theta\}$ of laws, indexed by a parameter space Θ.

Ex.

- $N(\theta, 1)$, $\Theta = \mathbb{R}$.
- $N(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$, $\Theta = \mathbb{R} \times (0, \infty)$.
- $\text{Poi}(\theta)$, $\Theta = [0, \infty)$.

Def 2. We say the model is correctly specified when $\exists\, \theta_0 \in \Theta$ such that $P = P_{\theta_0}$.
We will refer to the "true parameter" as θ_0.

Goals of Statistics

1. Estimation: We define an estimator
$$\widehat{\theta}_n = \widehat{\theta}_n(X_1, \ldots, X_n),$$
which we hope is close to θ_0 when $X_i \overset{\text{iid}}{\sim} P_{\theta_0}$, for all $\theta_0 \in \Theta$, when n is large.
2. Testing: We have a null hypothesis
$$H_0 : \theta \in \Theta_0 \subseteq \Theta,$$
and an alternative hypothesis
$$H_1 : \theta \in \Theta_1 \subseteq \Theta.$$
We define a test,
$$\psi_n := \psi_n(X_1, \ldots, X_n),$$
such that $\psi_n = 0$ if H_0 is true, and $\psi_n = 1$ if H_1 is true, w.h.p.

3. Uncertainty Quantification

- Define a confidence set $C_n(X_1, \ldots, X_n; \alpha)$, such that

$$P_\theta(\theta \in C_n) \overset{(\geq)}{=} 1 - \alpha \qquad \forall\, \theta \in \Theta,$$

  where α is the significance level.
- Define a Bayesian credible set, a region of high posterior probability.

I. LIKELIHOOD PRINCIPLE

Def 3. The likelihood function

$$L_n(\theta) = \prod_{i=1}^{n} f_\theta(X_i).$$

Note: This is a random function, as it depends implicitly on $X_1, \ldots, X_n$.

The log-likelihood function

$$\ell_n(\theta) = \sum_{i=1}^{n} \log f_\theta(X_i),$$

we write

$$\overline{\ell}_n(\theta) = \frac{1}{n}\ell_n(\theta)$$

for a normalised log-likelihood.

Def 4. A maximum likelihood estimator $\widehat{\theta}_{\text{MLE}}$ is any estimator with

$$L_n\big(\widehat{\theta}_{\text{MLE}}\big) = \max_{\theta \in \Theta} L_n(\theta).$$

Ex.

$$X_i \overset{\text{iid}}{\sim} \text{Poi}(\theta).$$

$$L_n(\theta) = \prod_{i=1}^{n} e^{-\theta} \frac{\theta^{X_i}}{X_i!}.$$

$$\ell_n(\theta) = -n\theta + (\log \theta) \sum X_i + \text{const.}$$

$$\ell_n'(\theta) = -n + \frac{\sum X_i}{\theta} \overset{\text{FOC}}{=} 0$$

$$\implies \quad \text{suggests } \widehat{\theta}_{\text{MLE}} = \frac{\sum_{i=1}^{n} X_i}{n} := \overline{X}_n \qquad \text{“sample mean”}.$$

SOC:

$$\ell_n''(\theta) = -\frac{1}{\theta^2} \sum X_i < 0 \qquad \text{if } \sum X_i > 0.$$

If $\sum X_i = 0$,

$$\ell_n(\theta) = -n\theta \quad \implies \quad \widehat{\theta}_{\text{MLE}} = 0.$$

In every case

$$\widehat{\theta}_{\text{MLE}} = \overline{X}_n.$$

Ex.

$$X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad \theta = (\mu, \sigma^2).$$

$$\ell_n(\theta) = -\frac{n}{2}\log\pi - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

FOC:

$$\frac{\partial \ell_n}{\partial \mu} = -\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu) = 0,$$

$$\frac{\partial \ell_n}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(X_i - \mu)^2 = 0.$$

We can check that the Hessian of ℓ_n is negative definite $\forall\, \theta \in \mathbb{R} \times (0, \infty)$.
Hence,

$$\widehat{\theta}_{\text{MLE}} = \begin{pmatrix} \overline{X}_n \\ \underbrace{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2}_{\text{“Sample variance”}} \end{pmatrix}.$$

Non-examinable

The likelihood principle poses that if two observations

$$X = (X_1, \ldots, X_n) \qquad \text{and} \qquad X^* = (X_1^*, \ldots, X_n^*)$$

give rise to likelihoods L_n, L_n^* respectively, such that

$$L_n(\theta) = L_n^*(\theta) \cdot M \qquad \forall\, \theta \in \Theta,$$

and some constant M, then they should produce the same “inference” on the true parameter.

- The MLE, Bayes estimators, credible sets satisfy the likelihood principle.
- Confidence sets, tests generally do not.

Birnbaum (1962)

$$\text{Conditionality} + \text{Sufficiency} \quad \Longrightarrow \quad \text{Likelihood principle.}$$

end of lecture 1

II. Information Geometry

The score function is

$$S_n(\theta) = \frac{\partial}{\partial \theta}\ell_n(\theta), \qquad \Theta \subseteq \mathbb{R}^d.$$

In many cases,

$$S_n(\widehat{\theta}_{\text{MLE}}) = 0,$$

where, by convention,

$$S_n(\theta') = \frac{\partial}{\partial \theta}\ell_n(\theta)\bigg|_{\theta=\theta'}.$$

Let X be a r.v. taking values in $\mathcal{X}$, and $g : \mathcal{X} \to \mathbb{R}$. Take a model $\{f(x,\theta) : \theta \in \Theta\}$. We will write

$$\mathbb{E}_\theta g(X) = \int g(x) f(x,\theta)\, dx, \qquad X \sim f(\cdot,\theta).$$

Obs 1.

Suppose the model $\{f(\cdot,\theta) : \theta \in \Theta\}$ is such that

$$\mathbb{E}_{\theta_0} \big|\log f(X,\theta)\big| < \infty \qquad \forall \theta \in \Theta.$$

Define

$$\ell(\theta) = \mathbb{E}_{\theta_0} \log f(X,\theta), \qquad X \sim P_{\theta_0}.$$

The normalised log-likelihood

$$\bar{\ell}_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} \log f(X_i,\theta)$$

can be thought of as a sample version of $\ell(\theta)$.

Suppose the model is correctly specified, such that X has pdf $f(\cdot,\theta_0)$ for some $\theta_0 \in \Theta$.

Fact. Then $\ell(\theta)$ is maximised at θ_0. This justifies using the MLE to approximate θ_0 when n is large.

Indeed,

$$\begin{aligned}
\ell(\theta) - \ell(\theta_0) &= \mathbb{E}_{\theta_0} \left(\log \frac{f(X,\theta)}{f(X,\theta_0)} \right) \\
&\le \log \mathbb{E}_{\theta_0} \left(\frac{f(X,\theta)}{f(X,\theta_0)} \right) \\
&= \log \int \frac{f(x,\theta)}{f(x,\theta_0)} f(x,\theta_0)\, dx \\
&= \log 1 = 0.
\end{aligned}$$

By Jensen's inequality, as log is concave, with equality if

$$\frac{f(X,\theta)}{f(X,\theta_0)}$$

is a.s. constant.

$$\therefore \quad \ell(\theta) \le \ell(\theta_0).$$

If

$$f(\cdot,\theta) = f(\cdot,\theta') \quad \Longrightarrow \quad \theta = \theta',$$

we say the parameter is identifiable.

In that case, if $\theta \neq \theta_0$, Jensen's inequality is strict, so θ_0 is actually the unique maximizer of ℓ.

Rmk. The difference

$$\ell(\theta_0) - \ell(\theta) = \int \log \frac{f(x,\theta_0)}{f(x,\theta)} f(x,\theta_0)\, dx$$

is called the Kullback–Leibler divergence or relative entropy between P_{θ_0} and P_θ.

A notion of distance which is not symmetric.

Obs 2.

$$0 = \frac{\partial}{\partial\theta} \underbrace{\int f(x,\theta)\,dx}_{=1\ \forall\theta} = \int \frac{\partial}{\partial\theta} f(x,\theta)\,dx$$
$$= \int \frac{\partial \log f(x,\theta)}{\partial\theta} f(x,\theta)\,dx$$
$$= \mathbb{E}_\theta\left[\frac{\partial \log f(X,\theta)}{\partial\theta}\right].$$

[There are some conditions for this. (Leibniz rule). See P & M.]
"$S_n(\theta)$ is centred when $X \sim P_\theta$."

Obs 3.

Def 5. Let $\Theta \subseteq \mathbb{R}^d$. For any $\theta \in \text{int}(\Theta)$, the Fisher Information matrix is

$$I(\theta) = \mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(X,\theta)\cdot\left(\frac{\partial}{\partial\theta}\log f(X,\theta)\right)^T\right], \qquad X \sim P_\theta.$$

Rmk. This is the covariance matrix of

$$S_n(\theta) = \frac{\partial}{\partial\theta}\log f(X,\theta)$$

when $X \sim P_\theta$.

Prop 1. For all $\theta \in \text{int}(\Theta)$, in a "regular" model,

$$I(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta\,\partial\theta^T}\log f(X,\theta)\right]$$

$$\left(I_{ij}(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta_i\,\partial\theta_j}\log f(X,\theta)\right]\right).$$

Proof.

$$0 = \frac{\partial^2}{\partial\theta\,\partial\theta^T}\int f(x,\theta)\,dx = \int \frac{\partial^2}{\partial\theta\,\partial\theta^T} f(x,\theta)\,dx.$$

[When this equality holds, we say the model is "regular".]
Now, we have

$$\frac{\partial^2}{\partial\theta\,\partial\theta^T}\log f(x,\theta) = \frac{\partial}{\partial\theta^T}\left[\frac{1}{f(x,\theta)}\frac{\partial}{\partial\theta}f(x,\theta)\right]$$
$$= -\frac{1}{f(x,\theta)^2}\frac{\partial}{\partial\theta}f(x,\theta)\left(\frac{\partial}{\partial\theta}f(x,\theta)\right)^T + \frac{1}{f(x,\theta)}\frac{\partial^2}{\partial\theta\,\partial\theta^T}f(x,\theta).$$

Take $-\mathbb{E}_\theta[\cdot]$ of both sides.

$$-\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta\,\partial\theta^T}\log f(X,\theta)\right]$$
$$= \mathbb{E}_\theta\left[\frac{1}{f(X,\theta)^2}\frac{\partial}{\partial\theta}f(X,\theta)\left(\frac{\partial}{\partial\theta}f(X,\theta)\right)^T\right] + 0$$
$$= \mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(X,\theta)\left(\frac{\partial}{\partial\theta}\log f(X,\theta)\right)^T\right] = I(\theta).$$

□

Tensorisation.

Suppose the observation X consists of n random variables $X_1, \ldots, X_n$. Then, the F.I. matrix is

$$I_n(\theta) = \mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n, \theta) \left(\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n, \theta) \right)^T \right].$$

If $X_1, \ldots, X_n$ are iid such that

$$f(X_1, \ldots, X_n, \theta) = \prod_{i=1}^{n} f(X_i, \theta),$$

where $\{f(x, \theta) : \theta \in \Theta\}$ is a model for X_1, then the F.I. is

$$\begin{aligned} I_n(\theta) &= \mathbb{E}_\theta \left[\sum_{i,j} \underbrace{\frac{\partial}{\partial \theta} \log f(X_i, \theta)}_{\substack{\text{mean 0 under} \\ P_\theta}} \left(\frac{\partial}{\partial \theta} \log f(X_j, \theta) \right)^T \right] \\ &= \mathbb{E}_\theta \left[\sum_{i=1}^{n} \frac{\partial}{\partial \theta} \log f(X_i, \theta) \left(\frac{\partial}{\partial \theta} \log f(X_i, \theta) \right)^T \right] \\ &= nI(\theta), \end{aligned}$$

where $I(\theta)$ is the Fisher Information for a single observation.

End of Lecture 2

Recap of Information Geometry

Observation 1.

$$\mathrm{KL}(P_{\theta_0}, P_\theta) = \mathbb{E}_{\theta_0} \left[\log \frac{f(X, \theta_0)}{f(X, \theta)} \right] \geq 0,$$

minimised at $\theta = \theta_0$.

Observation 2.

$$\mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} \log f(X, \theta) \right] = 0, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Observation 3.

$$I(\theta) := \mathrm{Var}_\theta \left[\frac{\partial}{\partial \theta} \log f(X, \theta) \right] = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta \, \partial \theta^T} \log f(X, \theta) \right], \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Why geometry? The F.I., $I(\theta)$, defines a metric tensor on the manifold

$$\{f(\cdot, \theta) : \theta \in \Theta\}.$$

This is known as the **Fisher–Rao metric**. KL divergence is approximated locally by this metric.

- **Reparametrisation invariance.** Take the Bernoulli(θ) model, where θ is the probability of success. Reparametrise it in terms of "log-odds":

$$\eta = \log\left(\frac{\theta}{1-\theta}\right) \in \mathbb{R}.$$

  The manifold

$$\{f(\cdot, \eta) : \eta \in \mathbb{R}\},$$

  equipped with the metric tensor $I(\eta)$, preserves geodesics, path lengths, etc.

- **Sufficient statistic invariance.**

Chentsov's theorems. These properties characterise the Fisher–Rao metric.

Proposition 2 (Cramér–Rao Lower Bound)

Let

$$\{f(\cdot, \theta) : \theta \in \Theta \subseteq \mathbb{R}\}$$

be a "regular" statistical model. Let $\tilde{\theta}(X_1, \ldots, X_n)$ be an unbiased estimator based on i.i.d. samples $X_1, \ldots, X_n$:

$$\mathbb{E}_\theta \tilde{\theta} = \theta, \qquad \forall \theta \in \Theta.$$

Then

$$\mathrm{Var}_\theta(\tilde{\theta}) \geq \frac{1}{nI(\theta)}, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Proof. Recall the Cauchy–Schwarz inequality for random variables Y, Z:

$$\big(\mathbb{E}[YZ]\big)^2 \leq \mathbb{E}[Y^2]\mathbb{E}[Z^2].$$

Let

$$Z = \tilde{\theta} - \theta, \qquad Y = \frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n; \theta).$$

Then

$$\mathbb{E}[Z^2] = \mathrm{Var}_\theta(\tilde{\theta}), \qquad \text{as } \tilde{\theta} \text{ is unbiased,}$$

and

$$\mathbb{E}[Y^2] = nI(\theta), \qquad \text{as the F.I. tensorises.}$$

Moreover,

$$\begin{aligned}
\mathbb{E}[YZ] &= \mathbb{E}_\theta\left[\tilde{\theta}\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n; \theta)\right] \\
&\quad - \theta \underbrace{\mathbb{E}_\theta\left[\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n; \theta)\right]}_{=0} \\
&= \int_{\mathcal{X}^n} \tilde{\theta}(x_1, \ldots, x_n) \frac{\partial f(x_1, \ldots, x_n; \theta)}{\partial \theta} \frac{f(x_1, \ldots, x_n; \theta)}{f(x_1, \ldots, x_n; \theta)}\, dx_1 \cdots dx_n \\
&= \frac{\partial}{\partial \theta} \underbrace{\int_{\mathcal{X}^n} \tilde{\theta}\, f(x_1, \ldots, x_n; \theta)\, dx_1 \cdots dx_n}_{=\mathbb{E}_\theta \tilde{\theta} = \theta} \qquad \text{(by regularity)} \\
&= \frac{\partial \theta}{\partial \theta} = 1.
\end{aligned}$$

Plug these expectations into the Cauchy–Schwarz inequality and rearrange. □

Remark. Sometimes, the MLE is unbiased and attains the Cramér–Rao lower bound, but not always. However, we will show this is true asymptotically.

If $\Theta \subseteq \mathbb{R}^p$ with $p > 1$, let

$$\Phi : \Theta \to \mathbb{R}$$

be a scalar function of θ. Consider an unbiased estimator $\tilde{\Phi}(X_1, \dots, X_n)$. Then one can show that

$$\mathrm{Var}_\theta(\tilde{\Phi}) \geq \frac{1}{n} \left(\frac{\partial \Phi(\theta)}{\partial \theta} \right)^T I(\theta)^{-1} \frac{\partial \Phi(\theta)}{\partial \theta}, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Corollary 1. If $\tilde{\theta}$ is not unbiased, we have

$$\mathrm{Var}_\theta(\tilde{\theta}) \geq \frac{\dfrac{\partial}{\partial \theta} \mathbb{E}_\theta \tilde{\theta}}{nI(\theta)}, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Example. Take $\Phi(\theta) = \alpha^T \theta$ for some $\alpha \in \mathbb{R}^p$. In this case,

$$\frac{\partial \Phi(\theta)}{\partial \theta} = \alpha,$$

and

$$\mathrm{Var}_\theta(\tilde{\Phi}) \geq \frac{1}{n} \alpha^T I(\theta)^{-1} \alpha.$$

Example Sheet 1

We observe a single pair

$$(X_1, X_2) \sim N\left(\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}, \Sigma \right),$$

where $\Sigma \succ 0$ is known. We wish to estimate θ_1.

Case 1. Assume that θ_2 is known. In this case, the model has one parameter and F.I. matrix

$$I_1(\theta_1).$$

Case 2. Assume that θ_2 is unknown. For any estimator

$$\tilde{\theta} = \begin{pmatrix} \tilde{\theta}_1 \\ \tilde{\theta}_2 \end{pmatrix},$$

we have

$$\tilde{\theta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T \tilde{\theta}.$$

So, if $\tilde{\theta}$ is unbiased, we have the Cramér–Rao lower bound

$$\mathrm{Var}_\theta(\tilde{\theta}_1) \geq \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T I(\theta)^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \left[I(\theta)^{-1} \right]_{11}.$$

Goal. Show

$$I_1(\theta_1)^{-1} < \left[I(\theta)^{-1} \right]_{11},$$

unless Σ is diagonal (X_1, X_2 independent).

II. Asymptotic Theory of the MLE

All reasonable estimators are asymptotically unbiased:

$$\mathbb{E}_\theta \tilde{\theta} \xrightarrow{n\to\infty} \theta, \qquad \forall \theta \in \Theta.$$

We wish to prove that estimators are **consistent**:

$$\tilde{\theta} \xrightarrow{?} \theta \qquad \text{as } n \to \infty \quad \text{under } P_\theta, \qquad \forall \theta \in \Theta.$$

For consistent estimators, a reasonable notion of optimality is **asymptotic efficiency**:

$$n \operatorname{Var}_\theta(\tilde{\theta}) \xrightarrow{n\to\infty} I(\theta)^{-1}, \qquad \forall \theta \in \operatorname{int}(\Theta).$$

We will show that the MLE is asymptotically efficient and, furthermore,

$$\hat{\theta}_{\text{MLE}} \approx N\left(\theta, n^{-1} I(\theta)^{-1}\right), \qquad \forall \theta.$$

This distributional convergence can be used for statistical inference.

Stochastic Convergence

Definition 1. Let $(X_n : n \in \mathbb{N})$ be a collection of random variables in some probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- We say that X_n converges almost surely to a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$, or

  $$X_n \xrightarrow{\text{a.s.}} X,$$

  if and only if

  $$\mathbb{P}\left(\left\{\omega : \|X_n(\omega) - X(\omega)\|_2 \xrightarrow{n\to\infty} 0\right\}\right) = 1.$$

- We say that X_n converges to X in probability, or

  $$X_n \xrightarrow{P} X,$$

  if, for every $\varepsilon > 0$,

  $$\mathbb{P}\left(\|X_n - X\|_2 > \varepsilon\right) \xrightarrow{n\to\infty} 0.$$

End of Lecture 3